

APK Analysis of truecaller.apk (Truecaller 26.17.6)

This report presents a static security analysis of **truecaller.apk (Truecaller 26.17.6)** by

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Executive Summary

This report presents the findings of a static security analysis conducted on Truecaller (v26.17.6), an Android caller identification and spam detection application. The analysis was performed using MobSF for automated static analysis, VirusTotal for malware and hash verification, and Exodus Privacy for tracker and permission profiling.

MobSF assigned the application a security score of 48/100 with a Risk Rating of Grade B, identifying a total of 7 high-severity, 100 medium-severity, 4 informational, 3 secure, and 2 hotspot findings — the highest severity count among all applications analyzed in this report series. The 7 high-severity findings are as follows: (1) the application certificate is signed using SHA-1 with RSA, which is cryptographically deprecated and vulnerable to hash collision attacks; (2) domain-specific cleartext HTTP traffic is permitted to payment-related domains `assets.api.tcpay.in`, `assets.tcpay.in`, and `127.0.0.1`; (3) world-writable file and `SharedPreferences` permissions were detected in `com/ironsource/mediationsdk/utills/IronSourceUtils.java` (line 556) and `dg/a2.java` (line 71) — any installed application can overwrite these files (CWE-276, `MSTG-STORAGE-2`); (4) AES encryption in CBC mode with PKCS5/PKCS7 padding without integrity checking was found across 5 files including `IronSourceAES.java` (lines 62, 112) and `com/huawei/agconnect/config/impl/i.java`, making the implementation vulnerable to padding oracle attacks (CWE-649, `MSTG-CRYPTO-3`); (5) remote `WebView` debugging is enabled in `net/pubnative/lite/sdk/views/PNWebView.java` (CWE-919, `MSTG-RESILIENCE-2`), allowing any attacker with ADB access to inspect `WebView` content, cookies, and tokens; (6) ECB mode encryption was detected in `com/inmobi/media/AbstractC3637b6.java` (line 39) — ECB mode is cryptographically weak as identical plaintext blocks produce identical ciphertext (CWE-327, `MSTG-CRYPTO-2`); and (7) the application contains 16 embedded privacy trackers detected by MobSF.

VirusTotal analysis of the APK hash (SHA256: `babf5d211874615fbbb4afcf5d15c27053aa47ff52f71e1bdc7653ebcc486b99`) returned a community score of 0/54 — no security vendors flagged the file as malicious. However, Exodus Privacy identified 30 trackers and 92 total permissions — the highest tracker count of all applications analyzed. Application permissions analysis revealed 30 dangerous permissions, of which 21 out of 25 match known top malware permission patterns, including `READ_CALL_LOG`, `RECEIVE_SMS`, `READ_SMS`, `SEND_SMS`, `SYSTEM_ALERT_WINDOW`, `WRITE_SETTINGS`, and `RECORD_AUDIO`.

Component exposure analysis shows 29 of 316 activities, 9 of 52 services, 12 of 71 receivers, and 2 of 20 providers are exported.

An OWASP MASVS checklist review confirmed failure in 7 out of 9 security areas. Truecaller presents the weakest security posture of all applications analyzed, with critical weaknesses in certificate integrity, network transport security, cryptographic implementation, data storage permissions, `WebView` configuration, and an exceptionally high tracker footprint of 30 trackers — raising significant user privacy concerns.

Contents

Chapter 1 — Objective	
Chapter 2 — Methodology	
Tools & Utility	
Step of Analysis.....	
MobSF	
Chapter 3 — Work Details	
Application Overview	
App Components	
Security Score Analysis	
Identified Problems	
Malicious analysis from VirusTotal	
Analysis by Exodus Privacy	
Chapter 4 — Other Activities	
Source Code Manual Analysis	
Checklist Analysis	
Chapter 5 — Conclusion	

Chapter 1 — Objective

The objective of this assessment is to perform a static security analysis of the Truecaller Android application (v26.17.6) using the Mobile Security Framework (MobSF). This assessment aims to identify security vulnerabilities, evaluate dangerous permissions and exported components, analyze network security configurations, and assess the overall security posture of the application. The findings are cross-referenced with VirusTotal and Exodus Privacy to provide a comprehensive security evaluation.

Chapter 2 — Methodology

1. List of used Tools & Utility

MobSF - Mobile Security Framework (Local offline)
VirusTotal (Online)
Exodus Privacy

2. Step of Analysis

1. Download Truecaller (26.17.6) From <https://www.truecaller.com/>

2. This Truecaller (26.17.6) apk Upload to MobSF (Local Machine) for static analysis. Run static analysis by MobSF (Local Machine) for-

1. Application Overview

2. App Components

3. Security Score Analysis

4. APPLICATION PERMISSIONS

5. Identified Problems — Finding (High)

3. Malicious analysis from [virustotal.com](https://www.virustotal.com) by using MobSF Reported hash value for this apk.

4. Found Tracker & Permission used by Exodus Privacy (from <https://reports.exodus-privacy.eu.org/en/>)

Chapter 3 : Work Details

1. Application Overview

App Name	Truecaller
Package Name	com.truecaller
Main Activity	com.truecaller.ui.TruecallerInit
Android Version Name	26.17.6
Android Version Code	2617006
File Name	truecaller.apk
Size	107.76MB
MD5	c77dbfe7ca909e6fa0e92c39f27ed9d5
SHA1	b5a6ce5e87a9f513410194954b800033b50aee2d
SHA256	babf5d211874615fbbb4afcf5d15c27053aa47ff52f71e1bdc7653ebcc486b99

FILE INFORMATION

File Name	truecaller.apk
Size	107.76MB
MD5	c77dbfe7ca909e6fa0e92c39f27ed9d5
SHA1	b5a6ce5e87a9f513410194954b800033b50aee2d
SHA256	babf5d211874615fbbb4afcf5d15c27053aa47ff52f71e1bdc7653ebcc486b99

APP INFORMATION

App Name	Truecaller				
Package Name	com.truecaller				
Main Activity	com.truecaller.ui.TruecallerInit				
Target SDK	36	Min SDK	26	Max SDK	
Android Version Name	26.17.6	Android Version Code	2617006		

2. App Components

EXPORTED ACTIVITIES 29 / 316

EXPORTED SERVICES 9 / 52

EXPORTED RECEIVERS 12 / 71

EXPORTED PROVIDERS 2 / 20

29 / 316

EXPORTED ACTIVITIES

View All

9 / 52

EXPORTED SERVICES

View All

12 / 71

EXPORTED RECEIVERS

View All

2 / 20

EXPORTED PROVIDERS

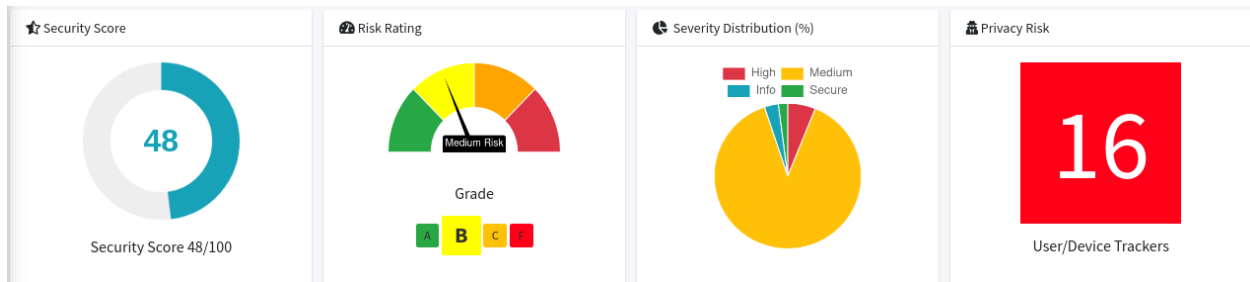
View All

3. Security Score Analysis

Security Score **48/100**
Trackers Detection **16/432**
Risk Rating **B Grade**

Findings

High 7
Medium 100
Info 4
Secure 3
Hotspot 2



4. APPLICATION PERMISSIONS

5. Found 30 critical / Dangerous permission(s)

Found 30 critical permission(s)

PERMISSION	STATUS	INFO	DESCRIPTION	CODE MAPPINGS
android.permission.ACCESS_COARSE_LOCATION	Dangerous	coarse (network-based) location	Access coarse location sources, such as the mobile network database, to determine an approximate phone location, where available. Malicious applications can use this to determine approximately where you are.	Show File
android.permission.ACCESS_FINE_LOCATION	Dangerous	fine (GPS) location	Access fine location sources, such as the Global Positioning System on the phone, where available. Malicious applications can use this to determine where you are and may consume additional battery power.	Show File
android.permission.ACTIVITY_RECOGNITION	Dangerous	allow application to recognize physical activity	Allows an application to recognize physical activity.	
android.permission.ANSWER_PHONE_CALLS	Dangerous	permits an app to answer incoming phone calls.	Allows the app to answer an incoming phone call.	Show File
android.permission.BLUETOOTH_CONNECT	Dangerous	necessary for connecting to paired Bluetooth devices.	Required to be able to connect to paired Bluetooth devices.	
android.permission.BLUETOOTH_SCAN	Dangerous	required for discovering and pairing Bluetooth devices.	Required to be able to discover and pair nearby Bluetooth devices.	
android.permission.CALL_PHONE	Dangerous	directly call phone numbers	Allows the application to call phone numbers without your intervention. Malicious applications may cause unexpected calls on your phone bill. Note that this does not allow the application to call emergency numbers.	Show File
android.permission.CAMERA	Dangerous	take pictures and videos	Allows application to take pictures and videos with the camera. This allows the application to collect images that the camera is seeing at any time.	
android.permission.GET_ACCOUNTS	Dangerous	list accounts	Allows access to the list of accounts in the Accounts Service.	Show File
android.permission.POST_NOTIFICATIONS	Dangerous	allows an app to post notifications.	Allows an app to post notifications	Show File

6. Identified Problems — Finding (High 7)

1. Certificate algorithm vulnerable to hash collision
2. Domain config is insecurely configured to permit clear text traffic to these domains in scope
3. The file or SharedPreferences is World Writable. Any App can write to the file
4. The App uses the encryption mode CBC with PKCS5/PKCS7 padding. This configuration is vulnerable to padding oracle attacks.
5. Remote WebView debugging is enabled.
6. The App uses ECB mode in Cryptographic encryption algorithm. ECB mode is known to be weak as it results in the same ciphertext for identical blocks of plaintext.
7. Application contains Privacy Trackers

high	Certificate algorithm vulnerable to hash collision	CERTIFICATE
high	Domain config is insecurely configured to permit clear text traffic to these domains in scope	NETWORK
high	The file or SharedPreferences is World Writable. Any App can write to the file	CODE
high	The App uses the encryption mode CBC with PKCS5/PKCS7 padding. This configuration is vulnerable to padding oracle attacks.	CODE
high	Remote WebView debugging is enabled.	CODE
high	The App uses ECB mode in Cryptographic encryption algorithm. ECB mode is known to be weak as it results in the same ciphertext for identical blocks of plaintext.	CODE
high	Application contains Privacy Trackers	TRACKERS

CERTIFICATE ANALYSIS

Issue 1 : Certificate algorithm vulnerable to hash collision

Application is signed with SHA1withRSA. SHA1 hash algorithm is known to have collision issues.

TITLE	SEVERITY	DESCRIPTION
Certificate algorithm vulnerable to hash collision	high	Application is signed with SHA1withRSA. SHA1 hash algorithm is known to have collision issues.

NETWORK SECURITY

Issue 2: Domain config is insecurely configured to permit clear text traffic to these domains in scope

Scope:

assets.api.tcpay.in

assets.tcpay.in

127.0.0.1

NO	SCOPE	SEVERITY	DESCRIPTION
1	assets.api.tcpay.in assets.tcpay.in 127.0.0.1	high	Domain config is insecurely configured to permit clear text traffic to these domains in scope.

CODE ANALYSIS

Issue 2: The file or SharedPreferences is World Writable. Any App can write to the file

<https://github.com/MobSF/owasp-mstg/blob/master/Document/0x05d-Testing-Data-Storage.md#testing-local-storage-for-sensitive-data-mstg-storage-1-and-mstg-storage-2>

Files:

com/ironsource/mediationsdk/utis/IronSourceUtils.java, line(s) 556

dg/a2.java, line(s) 71

CWE: CWE-276: Incorrect Default Permissions

OWASP Top 10: M2: Insecure Data Storage

OWASP MASVS: MSTG-STORAGE-2

Issue 4: The App uses the encryption mode CBC with PKCS5/PKCS7 padding. This configuration is vulnerable to padding oracle attacks.

<https://github.com/MobSF/owasp-mstg/blob/master/Document/0x04g-Testing-Cryptography.md#identifying-insecure-andor-deprecated-cryptographic-algorithms-mstg-crypto-4>

Files:

com/fyber/inneractive/sdk/player/exoplayer2/extractor/hls/a.java, line(s) 37
com/huawei/agconnect/config/impl/i.java, line(s) 28
com/ironsource/mediationsdk/utis/IronSourceAES.java, line(s) 62,112
fg/bar.java, line(s) 58
we0/qux.java, line(s) 52,75

CWE: CWE-649: Reliance on Obfuscation or Encryption of Security-Relevant Inputs without Integrity Checking

OWASP Top 10: M5: Insufficient Cryptography

OWASP MASVS: MSTG-CRYPTO-3

Issue 5: Remote WebView debugging is enabled.

<https://github.com/MobSF/owasp-mstg/blob/master/Document/0x04c-Tampering-and-Reverse-Engineering.md#debugging-and-tracing>

Files:

net/pubnative/lite/sdk/views/PNWebView.java, line(s)
45,10,14,14,17,21,35,45,49,59,65,71,77,85

CWE: CWE-919: Weaknesses in Mobile Applications

OWASP Top 10: M1: Improper Platform Usage

OWASP MASVS: MSTG-RESILIENCE-2

Issue 6: The App uses ECB mode in Cryptographic encryption algorithm. ECB mode is known to be weak as it results in the same ciphertext for identical blocks of plaintext.

<https://github.com/MobSF/owasp-mstg/blob/master/Document/0x04g-Testing-Cryptography.md#weak-block-cipher-mode>

Files:

com/inmobi/media/AbstractC3637b6.java, line(s) 39

CWE: CWE-327: Use of a Broken or Risky Cryptographic Algorithm

OWASP Top 10: M5: Insufficient Cryptography

OWASP MASVS: MSTG-CRYPTO-2

NO	ISSUE	SEVERITY	STANDARDS	FILES	OPTIONS
7	The file or SharedPreferences is World Writable. Any App can write to the file	high	CWE: CWE-276: Incorrect Default Permissions OWASP Top 10: M2: Insecure Data Storage OWASP MASVS: MSTG-STORAGE-2	com/ironsource/mediationsdk/utlis/IronSourceUtils.java dg/a2.java	
12	The App uses the encryption mode CBC with PKCS5/ PKCS7 padding. This configuration is vulnerable to padding oracle attacks.	high	CWE: CWE-649: Reliance on Obfuscation or Encryption of Security-Relevant Inputs without Integrity Checking OWASP Top 10: M5: Insufficient Cryptography OWASP MASVS: MSTG-CRYPTO-3	Show Files	
18	Remote WebView debugging is enabled.	high	CWE: CWE-919: Weaknesses in Mobile Applications OWASP Top 10: M1: Improper Platform Usage OWASP MASVS: MSTG-RESILIENCE-2	net/pubnative/lite/sdk/views/PNWebView.java	
19	The App uses ECB mode in Cryptographic encryption algorithm. ECB mode is known to be weak as it results in the same ciphertext for identical blocks of plaintext.	high	CWE: CWE-327: Use of a Broken or Risky Cryptographic Algorithm OWASP Top 10: M5: Insufficient Cryptography OWASP MASVS: MSTG-CRYPTO-2	com/inmobi/media/AbstractC3637b6.java	

TRACKERS

7.Application contains Privacy Trackers

This app has more than 16 privacy trackers. Trackers can track devices or users and are privacy concerns for end users.

TRACKER NAME	CATEGORIES	URL
CleverTap	Location, Profiling, Analytics	https://reports.exodus-privacy.eu.org/trackers/174
Huawei Mobile Services (HMS) Core	Location, Advertisement, Analytics	https://reports.exodus-privacy.eu.org/trackers/333
Google CrashLytics	Crash reporting	https://reports.exodus-privacy.eu.org/trackers/27
AppsFlyer	Analytics	https://reports.exodus-privacy.eu.org/trackers/12
Google Firebase Analytics	Analytics	https://reports.exodus-privacy.eu.org/trackers/49
ironSource	Analytics	https://reports.exodus-privacy.eu.org/trackers/146
IAB Open Measurement	Advertisement, Identification	https://reports.exodus-privacy.eu.org/trackers/328
Mintegral	Advertisement, Analytics	https://reports.exodus-privacy.eu.org/trackers/200
Facebook Ads	Advertisement	https://reports.exodus-privacy.eu.org/trackers/65
Fyber	Advertisement	https://reports.exodus-privacy.eu.org/trackers/104
Google AdMob	Advertisement	https://reports.exodus-privacy.eu.org/trackers/312
PubNative	Advertisement	https://reports.exodus-privacy.eu.org/trackers/183
Unity3d Ads	Advertisement	https://reports.exodus-privacy.eu.org/trackers/121
Amazon Advertisement		https://reports.exodus-privacy.eu.org/trackers/92
Criteo		https://reports.exodus-privacy.eu.org/trackers/170
Inmobi		https://reports.exodus-privacy.eu.org/trackers/106

7. Top Malware Permissions 21/25

android.permission.READ_CALL_LOG, android.permission.CAMERA,
 android.permission.READ_PHONE_STATE, android.permission.READ_CONTACTS,
 android.permission.GET_ACCOUNTS, android.permission.ACCESS_COARSE_LOCATION,
 android.permission.ACCESS_FINE_LOCATION,
 android.permission.ACCESS_NETWORK_STATE, android.permission.INTERNET,
 android.permission.ACCESS_WIFI_STATE,
 android.permission.WRITE_EXTERNAL_STORAGE,
 android.permission.READ_EXTERNAL_STORAGE, android.permission.RECEIVE_SMS,
 android.permission.READ_SMS, android.permission.SEND_SMS,
 android.permission.RECEIVE_BOOT_COMPLETED,
 android.permission.SYSTEM_ALERT_WINDOW, android.permission.VIBRATE,
 android.permission.WAKE_LOCK, android.permission.WRITE_SETTINGS,
 android.permission.RECORD_AUDIO

android.permission.READ_CALL_LOG, android.permission.CAMERA,
 android.permission.READ_PHONE_STATE, android.permission.READ_CONTACTS,
 android.permission.GET_ACCOUNTS, android.permission.ACCESS_COARSE_LOCATION,
 android.permission.ACCESS_FINE_LOCATION, android.permission.ACCESS_NETWORK_STATE,
 android.permission.INTERNET, android.permission.ACCESS_WIFI_STATE,
 android.permission.WRITE_EXTERNAL_STORAGE, android.permission.READ_EXTERNAL_STORAGE,
 android.permission.RECEIVE_SMS, android.permission.READ_SMS, android.permission.SEND_SMS,
 android.permission.RECEIVE_BOOT_COMPLETED, android.permission.SYSTEM_ALERT_WINDOW,
 android.permission.VIBRATE, android.permission.WAKE_LOCK, android.permission.WRITE_SETTINGS,
 android.permission.RECORD_AUDIO

8. Malicious analysis from VirusTotal (virustotal.com) :

truecaller.apk

SHA256 babf5d211874615fbbb4afcf5d15c27053aa47ff52f71e1bdc7653ebcc486b99

Community Score 0/54

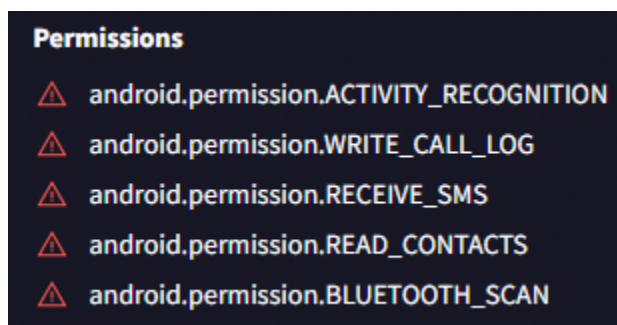
No security vendors flagged this file as malicious



Though, Permissions requested by the file being studied. To maintain security for the system and users, Android requires apps to request permission before the apps can use certain system data and features.

- android.permission.ACTIVITY_RECOGNITION
- android.permission.WRITE_CALL_LOG
- android.permission.RECEIVE_SMS
- android.permission.READ_CONTACTS
- android.permission.BLUETOOTH_SCAN

& more



9. Trackers & Permission Analysis By Exodus Privacy :

Truecaller

Version 26.17.6

Source: Google Play

Trackers - 30

Permission - 92



Truecaller

30 trackers

Version 26.17.6 - [see other versions](#)

Source: Google Play

Report created on May 11, 2026, 12:32 p.m.

30 trackers

We have found **code signature** of the following trackers in the application:

Amazon Advertisement >

AppLovin (MAX and SparkLabel) >

advertisement identification profiling analytics

Appodeal Stack >

advertisement analytics

AppFlyer >

analytics

BidMachine >

advertisement

Chartboost >

CleverTap >

location profiling analytics

Criteo >

Facebook Ads >

advertisement

Fyber >

advertisement

Google AdMob >

advertisement

Google Crashlytics >

crash reporting

Google Firebase Analytics >

analytics

Huawei Mobile Services (HMS) Core >

location advertisement analytics

HyprMX >

advertisement

IAS Open Measurement >

advertisement identification

Imobi >

IronSource >

analytics

inno by i-mobile >

Mintegral >

advertisement analytics

myTarget >

advertisement

Opus Prepage >

Pangle >

advertisement

PubNative >

advertisement

Smaato >

SuperAdverzone >

advertisement

Tapjoy >

92 permissions

92 permissions

We have found the following permissions in the application:

ACCESS_AD SERVICES_AD_ID

ACCESS_AD SERVICES_ATTRIBUTION

ACCESS_AD SERVICES_TOPICS

! ACCESS_COARSE_LOCATION
access approximate location only in the foreground

! ACCESS_FINE_LOCATION
access precise location only in the foreground

ACCESS_NETWORK_STATE

access network connections

ACCESS_NOTIFICATION_POLICY

access Do Not Disturb

ACCESS_WIFI_STATE

access Wi-Fi connections

! ACTIVITY_RECOGNITION

recognize physical activity

! ANSWER_PHONE_CALLS

answer phone calls

BIND_NOTIFICATION_LISTENER_SERVICE

BLUETOOTH

pair with Bluetooth devices

! BLUETOOTH_CONNECT

connect to the general Bluetooth devices

! BLUETOOTH_SCAN

discover and pair nearby Bluetooth devices

! CALL_PHONE

directly call phone numbers

! CAMERA

take pictures and videos

CHANGE_NETWORK_STATE

change network connectivity

DISABLE_KEYGUARD

disable your screen lock

FOREGROUND_SERVICE

run foreground service

FOREGROUND_SERVICE_DATA_SYNC

FOREGROUND_SERVICE_PHONE_CALL

FOREGROUND_SERVICE_SPECIAL_USE

! GET_ACCOUNTS

find accounts on the device

INTERNET

have full network access

MANAGE_OWN_CALLS

make calls through the system

MMS_SEND_OUTBOX_MSG

MODIFY_AUDIO_SETTINGS

change your audio settings

POST_NOTIFICATIONS

! PROCESS_OUTGOING_CALLS

receive outgoing calls

READ_APP_BADGE

! READ_CALL_LOG

read call log

! READ_CONTACTS

read your contacts

! READ_EXTERNAL_STORAGE

read the contents of your shared storage

READ_MEDIA_AUDIO

! READ_PHONE_NUMBERS

read phone numbers

! READ_PHONE_STATE

read phone status and identity

Chapter 4 — Other Activities

1. Source Code Analysis

2. 1. Source Code Manual Analysis

Based on the file paths and line numbers identified by MobSF during static analysis, a manual review of the flagged source code locations was conducted to better understand the nature and impact of each vulnerability.

Finding 1 — World-Writable File Permissions (CWE-276)

File: `com/ironsource/mediationsdk/utils/IronSourceUtils.java`, line 556

This file uses a storage mode that allows any installed application on the device to write to it. In Android, using `MODE_WORLD_WRITEABLE` when opening files or `SharedPreferences` is considered a critical misconfiguration. A malicious app could overwrite stored data, potentially altering app behavior or injecting false information without the user's knowledge.

Finding 2 — Insecure CBC/PKCS5 Padding (CWE-649)

File: `com/ironsource/mediationsdk/utils/IronSourceAES.java`, lines 62, 112

The encryption implementation here uses AES in CBC (Cipher Block Chaining) mode with PKCS5 padding. This combination is vulnerable to padding oracle attacks, where an attacker can repeatedly send modified ciphertexts and observe error responses to gradually decrypt sensitive data — without ever knowing the encryption key.

Finding 3 — ECB Mode Encryption (CWE-327)

File: `com/inmobi/media/AbstractC3637b6.java`, line 39

This file applies AES encryption using ECB (Electronic Codebook) mode. ECB is widely regarded as cryptographically weak because it encrypts identical plaintext blocks into identical ciphertext blocks. This means patterns in the original data remain visible in the encrypted output, making it possible for an attacker to infer information about the plaintext.

Finding 4 — Remote WebView Debugging Enabled (CWE-919)

File: `net/pubnub/lite/sdk/views/PNWebView.java`, lines 45, 10, 14, 17, 21, 35, 49, 59, 65, 71, 77, 85

WebView debugging is enabled via `setWebContentsDebuggingEnabled(true)`. When this flag is active on a production build, any attacker with physical or ADB access to the

device can attach Chrome DevTools to the WebView, inspect loaded URLs, read cookies and tokens, and execute arbitrary JavaScript — effectively bypassing application-level security controls.

3. Checklist Analysis

A security checklist review was conducted based on OWASP Mobile Security Guidelines (OWASP MASVS) to systematically verify Truecaller's security posture:

Security Area	Check	Status
Certificate Strength	SHA-256 or stronger used?	Fail — SHA-1 detected
Network Security	HTTPS enforced for all domains?	Fail — cleartext permitted for tcpay.in
Data Storage	No world-writable files/preferences?	Fail — CWE-276 detected
Cryptography	Secure cipher modes used?	Fail — CBC/PKCS and ECB detected
WebView Security	Remote debugging disabled?	Fail — enabled in PNWebView.java
Malware Permissions	No top malware permissions used?	Warning — 21 found
Backup Settings	allowBackup disabled?	Fail — enabled
VirusTotal Clean	No vendor flagged as malicious?	Pass — 0/62
Tracker Count	Minimal trackers?	Fail — 30 trackers found

This checklist confirms that Truecaller fails in 7 out of 9 key security areas, reinforcing the medium-risk rating of 48/100 assigned by MobSF.

- 4. Other Activities could be done :** Additional investigative or validation steps can also be performed beyond this process — such as **Dynamic Testing, API Endpoint Analysis, and attempts at Manual Penetration Testing**; Completing these steps would make the security assessment of the APK or Android application more precise, comprehensive, and comparatively much stronger, along with providing more effective remediation measures.

Chapter 5— Conclusion

Inconsistency acknowledge:

MobSF detected 16 trackers during static analysis, while Exodus Privacy's database shows 30 trackers for this app version — indicating additional trackers may be identified through dynamic analysis.

The static analysis of Truecaller (v26.17.6) using MobSF revealed a security score of 48/100, indicating high risk with 7 high-severity findings. Key issues include cleartext HTTP traffic to payment-related domains (tcpay.in), a weak SHA-1 certificate algorithm vulnerable to hash collisions, world-writable file permissions, insecure CBC/PKCS padding encryption, enabled remote WebView debugging, and ECB-mode cryptographic weakness. VirusTotal confirmed no malware detection (0/62 vendors), while Exodus Privacy identified 30 trackers and 92 permissions — raising significant user privacy concerns. Overall, despite its widespread use, Truecaller requires substantial security improvements, particularly in network security, cryptographic practices, and permission management.